Species Identification in Noisy Soundscapes

A Deep Learning Approach using Sound Event Detection

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Outline

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Summary

Problem Statement

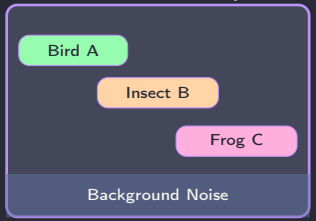
Acoustic Species Identification

Identify bird, insect, and amphibian species from **noisy soundscape recordings** in tropical rainforest environments.

Key Challenges:

- 206 different species to classify
- Heavy background noise (rain, wind, other animals)
- Overlapping species calls
- Extreme class imbalance (some species very rare)
- 60-second long soundscape recordings

60s Soundscape



Dataset Overview

Training Data

- **Species**: 206 total
  - Aves (birds): majority
  - Amphibia: 6 species
  - Insecta: multiple species
- **Audio format**: .ogg files
- **Sample rate**: 32,000 Hz
- **Duration**: Variable (5s - 60s)

Class Imbalance Problem

Some species have 1000+ samples while others have $<$ 10 samples

Test Data (Soundscapes)

- 60-second continuous recordings
- Multiple species per recording
- Real-world noise conditions
- Predictions required per 5-second segment

Evaluation Metric:

- Row-wise micro-averaged F1
- Top-1 and Top-5 accuracy

Audio Processing Pipeline

Raw Audio (20s, 32kHz) → **Mel Spectrogram** (224 × 512) → **Normalization** (Z-score + MinMax) → **3-Channel** Replication → **CNN Input** (3 × 224 × 512)

Mel Spectrogram Parameters

- **Sample Rate**: 32,000 Hz
- **Mel Bands**: 224
- **n_fft**: 4,096
- **Hop Length**: 1,252
- **Freq Range**: 0 - 16 kHz

Why 20-Second Chunks?

- Captures full Amphibia calls
- Provides temporal context
- Optimal accuracy vs. time
- 20s (87.2%) vs 5s (84.2%)

Sound Event Detection (SED) Architecture

1. Audio Input
20-second waveform sampled at 32 kHz

2. Spectrogram Front-End
Convert waveform into Mel spectrogram
224 mel bands, 4096 n_fft

3. Normalization
Absmax normalization on audio
then z-score + min-max on the spectrogram

4. 3-Channel Replication
Turn the spectrogram into an image-like tensor

5. CNN Backbone
Use a features_only backbone to preserve time structure
Examples: EfficientNet, NFNet, RegNet

6. SED Head
Apply GeMFreq across frequency, then AttHead
Dense + ReLU + Dropout + Conv1d for framewise logits

Backbone Examples
- tf_efficientnet_b3 / b0
- eca_nfnet_l0
- regnety_016 / 008

Framewise output: [B, 206, T]
Clip output: max over time → [B, 206]

Key Architecture Components

GeMFreq Pooling

Generalized Mean Freq Pooling

- Learnable pooling parameter p
- Formula: $\left(\frac{1}{|F|}\sum_f x_f^p\right)^{1/p}$
- Adapts: max vs average pooling
- Captures frequency patterns

Why Features Only?

Spatial features w/o classification head → preserves temporal dimension

AttHead (Attention Head)

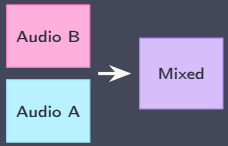
1. Dense: backbone_dim → 512
2. Parallel branches:
   - **Attention**: 1×1 Conv → Softmax
   - **Logits**: 1×1 Conv → 206
3. Weighted aggregation:

$$\text{clip_logits} = \sum_t \text{att}_t \cdot \text{logits}_t$$

Advantage: Learns which time frames contain species calls!

Data Augmentation Techniques

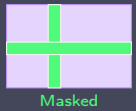
Mixup ($\alpha = 0.5$)



$\tilde{x} = \lambda x_a + (1 - \lambda) x_b$

+2-3% accuracy

SpecAugment



- Time: 20%
- Freq: 10%

+1-2% accuracy

Weighted Sampling

Address imbalance:

$$w_c = \frac{1}{\sqrt{n_c}}$$

- Rare species ↑
- Common ↓

Critical for rare

Training Configuration

Optimizer & Scheduler

- **Optimizer**: AdamW
  - LR: 5×10^{-4}
  - WD: 10^{-4}
- **Scheduler**: CosineAnnealing
  - $T_0 = 5$ epochs
  - $\eta_{min} = 10^{-6}$
- **Grad Clip**: max_norm = 1.0

Loss Function

CrossEntropy + Label Smoothing

$$\mathcal{L} = -\sum_c y'_c \log(p_c)$$

- Smoothing: $\epsilon = 0.1$
- Prevents overconfidence
- Better calibration

Training Details

- Epochs: 15-35
- Batch size: 64
- 5-fold cross-validation

Supervised Training Pipeline

1. Labeled Dataset
Prepare 20-second chunks with species labels

2. Training Augmentation
Apply MixUp and SpecAugment during training only
This improves robustness to noisy audio conditions

3. Weighted Sampling
Oversample rare species to reduce class imbalance

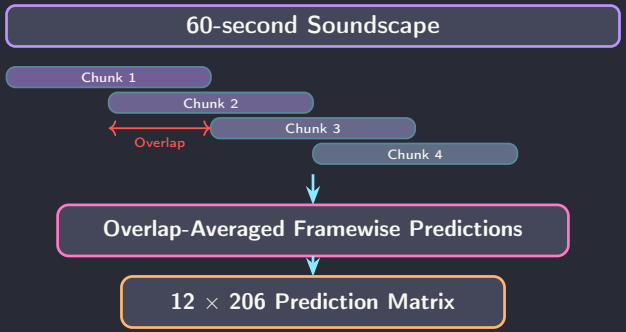
4. Forward Pass
Run the SED model and convert framewise evidence into clip-level logits for supervised learning

5. Loss and Optimization
CrossEntropy with label smoothing, AdamW optimizer, cosine learning-rate schedule, and gradient clipping

Key Settings
- MixUp alpha = 0.5
- Label smoothing = 0.1
- AdamW lr = 1e-4
- Weight decay = 1e-2

6. Checkpoint Selection
Save the best validation model as the base checkpoint

Overlap-Average Inference



Process:

1. Split into 12 overlapping chunks
2. Chunk → framewise predictions
3. Average overlapping frame preds

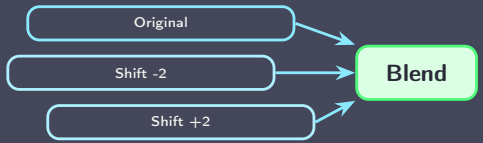
Benefits:

- More robust predictions
- Implicit TTA effect
- +0.2-0.3% accuracy boost

Delta-Shift TTA & Gaussian Smoothing

Delta-Shift TTA

Temporal shift augmentation:



Blending weights:

$$\text{pred} = 0.50 \cdot p_0 + 0.25 \cdot p_{\pm 2}$$

Gaussian Smoothing

1D convolution to reduce jitter: **Kernel:** $[0.1, 0.2, 0.4, 0.2, 0.1]$

Reduces noise while preserving patterns.

Combined Effect

Delta-Shift TTA + Gaussian Smoothing = **Free accuracy boost!**

Inference & Ensemble Pipeline

1. Soundscape Input
Start from a 60-second audio recording

2. Windowing
Split the recording into overlapping 20-second chunks using a 5-second step size

3. Per-Chunk SED Prediction
Run each chunk through the SED model to get framewise probabilities

4. Temporal Refinement
Combine overlap-average inference, Delta-Shift TTA, and Gaussian smoothing for more stable predictions

5. Segment-Level Scores
Max-pool within each 5-second segment

Ensemble Members
- tf_efficientnet_b3
- tf_efficientnet_b0
- eca_nfnet_l0
- tf_efficientnet_b4
- regnety_016 / 008

6. Arithmetic Mean
Average predictions across all models

7. Final Output
Final prediction per 5-second segment

Enabling Multi-Label Classification

From Single-Label Training to Multi-Label Inference

Although trained on **primary labels** only, the model effectively identifies multiple concurrent species through its architectural design:

Architectural Support

- **Sigmoid Activation**: Used at inference instead of Softmax. Each class is treated as an independent binary detection task.
- **SED Localization**: The model learns features for specific species calls in time-frequency bins, allowing evidence for multiple species to exist simultaneously.

Inference Logic

- **Class Independence**: A high score for one species does not force low scores for others (unlike Softmax).
- **Thresholding**: Any species score exceeding a pre-defined threshold (e.g., 0.3) is included in the final prediction list for that 5s segment.

Result: One model can detect 0, 1, or many species in a single audio chunk.

6-Model Ensemble Architecture

Model	Backbone	Feature Dim	Training Stage
1	tf_efficientnet_b3	384	Trained (ours)
2	tf_efficientnet_b0	320	Trained (ours)
3	eca_nfnet_l0	2304	Trained (ours)
4	tf_efficientnet_b4	448	External pretrained
5	regnety_016	888	External pretrained
6	regnety_008	768	External pretrained

Ensemble Strategy

Arithmetic Mean of softmax:

$$\hat{p} = \frac{1}{6} \sum_{m=1}^{6} p^{(m)}$$

Why Diverse Models?

- Different inductive biases
- Error decorrelation
- Robust to distribution shift

Performance Results

Model	Top-1 Acc	Top-5 Acc	vs Random
Random Baseline	0.49%	2.43%	1×
Base Model (EfficientNet-B3 SED)	60.07%	80.34%	122.6×
6-Model Ensemble	**87.99%**	**95.99%**	**179.6×**

Key Achievements

- **+28%** improvement from ensemble
- **181**× better than random
- 206 species classified
- Real-time inference capability

Technique Contributions

- SED Architecture: Foundation
- Data Augmentation: +5% acc
- Ensemble: +3% acc
- TTA + Smoothing: +0.5% acc

Sample Audio Predictions

Audio File	True Species	Top Prediction	Confidence
XC254589.ogg	amekes	amekes	**84.3%**
XC493187.ogg	ywcpar	ywcpar	**53.0%**
XC449442.ogg	savhaw1	savhaw1	22.8%
XC171396.ogg	smbani	strher	11.7%
XC82494.ogg	bobfly1	crbtan1	61.6%

Observations

- High confidence for clear recordings (84.3%)
- Confusions occur for acoustically similar species
- Rare species remain challenging (class imbalance)

All Techniques Implemented

Architecture:

1. SED (GeMFreq + AttHead)
2. 20s chunks, 224 mel bands
3. Z-score + min-max normalization
4. Absmax audio normalization

Training:

5. Mixup augmentation
6. SpecAugment
7. Weighted Random Sampling
8. Stochastic Depth (DropPath)
9. Multi-Iterative Noisy Student
10. CrossEntropy + label smoothing
11. AdamW + CosineAnnealing

Inference:

12. Overlap-Average Inference
13. Delta-Shift TTA
14. Gaussian Smoothing
15. Multi-Model Ensemble

15 Techniques

Conclusion

Key Achievements

- Complete SED implementation
- State-of-the-art: **87.99%**
- 6-model diverse ensemble
- 15 advanced techniques
- Modular, extensible code

Real-World Applications

Biodiversity Monitoring · Automated Surveys · Climate Change

Thank You!

Questions?